Discrete Structures Assignment # 2 Date

Q1) (i)

$f(x) = \left\lfloor \frac{x^2}{3} \right\rfloor$ $\quad f(-2) = \left\lfloor \frac{4}{3} \right\rfloor = 1$ $\quad f(0) = \left\lfloor \frac{0}{3} \right\rfloor = 0$

$S = \{-2, -1, 0, 1, 2, 3\}$ $\quad f(-1) = \left\lfloor \frac{1}{3} \right\rfloor = 0$ $\quad f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0$

$f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1$ $\quad f(3) = \left\lfloor \frac{9}{3} \right\rfloor = 3$

$\boxed{f(S) = \{0, 1, 3\}}$

(iii) $S = \{1, 5, 7, 11\}$

$f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0$ $\quad f(5) = \left\lfloor \frac{25}{3} \right\rfloor = 8$ $\quad f(7) = \left\lfloor \frac{49}{3} \right\rfloor = 16$

$f(11) = \left\lfloor \frac{121}{3} \right\rfloor = 40$

$\boxed{f(S) = \{0, 8, 16, 40\}}$

(ii) $S = \{0, 1, 2, 3, 4, 5\}$

$f(0) = 0$ $\quad f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0$ $\quad f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1$ $\quad f(3) = \left\lfloor \frac{9}{3} \right\rfloor = 3$

$f(4) = \left\lfloor \frac{16}{3} \right\rfloor = 5$ $\quad f(5) = \left\lfloor \frac{25}{3} \right\rfloor = 8$

$\boxed{f(S) = \{0, 1, 3, 5, 8\}}$

(iv) $S = \{2, 6, 10, 14\}$

$f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1$ $\quad f(14) = \left\lfloor \frac{196}{3} \right\rfloor = 65$

$f(6) = \left\lfloor \frac{36}{3} \right\rfloor = 12$

$f(10) = \left\lfloor \frac{100}{3} \right\rfloor = 33$

$\boxed{f(S) = \{1, 12, 33, 65\}}$

Q2) $\lceil -x \rceil = -\lceil x \rceil$

$\lceil -x \rceil = -\lceil x \rceil$ → (ii) Again same thing so disproved again

(i)

$\lceil -x \rceil = -\lceil x \rceil$

Let $x = 1/2$ then

$\left\lceil -\frac{1}{2} \right\rceil = -1$

but $-\left\lceil \frac{1}{2} \right\rceil = 0$

$-1 \neq 0$ so disproved

Q3) $f(a) = 2a + 3$ $g(a) = 3a + 2$

(a) f of g, g of f

$(f \circ g)a = f(g(a))$

$2(3a+2) + 3 = 6a + 4 + 3 = 6a + 7$

$(g \circ f)a = g(f(a))$

$= 3(2a+3) + 2 = 6a + 9 + 2$

$= 6a + 11$

(b)

$f(a) = 2a + 3$ Suppose:

$f(a_1) = f(a_2)$

$2a_1 + 3 = 2a_2 + 3$

$a_1 = a_2$

f is one-to-one

but $f(a) = 2a + 3$ produces odd integer too so it's not onto.

Suppose: $g(a_1) = g(a_2)$

$3a_1 + 2 = 3a_2 + 2$

$a_1 = a_2$

g is one-to-one but not onto.

Both f and g are one-to-one but not onto

(c)

No

Q4) (a) $a = b$

$R = \{(0,0), (1,1), (2,2), (3,3)\}$

(b) $a + b = 4$

$R = \{(1,3), (2,2), (3,1), (4,0)\}$

(c) $a > b$

$R = \{(1,0), (2,0), (2,1), (3,0), (3,1), (3,2), (4,0), (4,1), (4,2), (4,3)\}$

(d) $a \mid b-a$ divides b

$R = \{(1,0), (1,1), (1,2), (1,3), (2,0), (2,2), (3,0), (3,3)\}$

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(e) $\gcd(a,b) = 1$

$R = \{(0,1), (1,0), (1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (4,1), (4,3)\}$

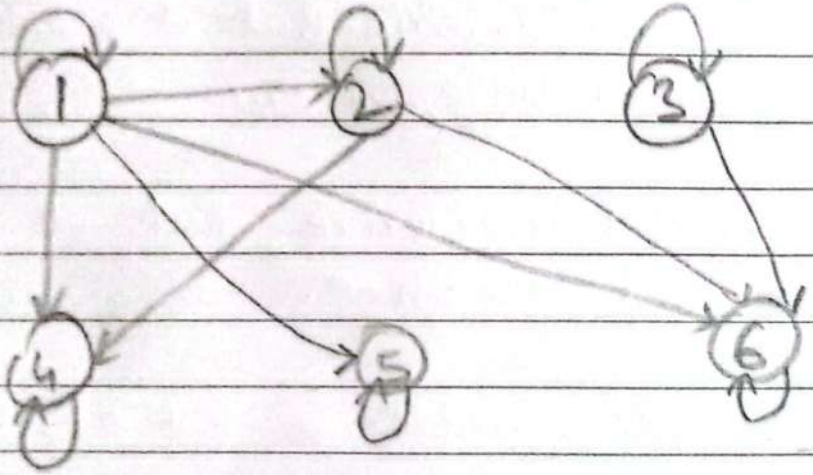
(f) $\text{lcm}(a,b) = 2$

$R = \{(1,2), (2,1), (2,2)\}$

Q5) (a) Ordered pairs

$R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\}$

(b) Directed Graph



Matrix Form: $M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

- Relation R is a partial order relation not equivalence.

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Q6)(a) ~~Reflexive~~ Not reflexive, Not symmetric, Not antisymmetric but it's Transitive.

(b)
It's reflexive, symmetric, transitive but not antisymmetric

(c)
It's symmetric and antisymmetric but not reflexive or transitive

(d)
It's antisymmetric but not reflexive, symmetric or transitive

(e)
It's all.

(f)
It's not anything.

Q7)(a) Not reflexive or symmetric. But it's antisymmetric, irreflexive and transitive asymmetric.

(b)
Not antisymmetric, asymmetric, irreflexive but it is reflexive, symmetric and transitive.

(c)
Not antisymmetric, asymmetric, irreflexive but it is reflexive, symmetric and transitive.

(d)
~~Ref~~ Not antisymmetric, asymmetric, irreflexive or transitive but it is reflexive and symmetric.

Q8)(a)
$R = \{(0,0), (1,1)\}$

(b)
$R = \{(0,1), (1,0), (1,2)\}$

(c)
$R = \{(0,0), (1,1)\}$

(d)
$R = \{(0,1), (1,2)\}$

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Q9) (a) $R_2 \cup R_4$

$\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

(b) $R_3 \cup R_6$

$\{(1,2), (1,3), (2,1), (2,3), (3,1), (3,2)\}$

(c) $R_3 \cap R_6$

$\{(1,2), (1,3), (2,3)\}$

(d) $R_4 \cap R_6$

$\{(1,2), (1,3), (2,3)\}$

(e) $R_3 - R_6$

$\emptyset$

(f) $R_6 - R_3$

$\{(2,1), (3,1), (3,2)\}$

(g) $R_2 \oplus R_6$

~~$\{(1,1), (2,2), (3,3)\}$~~

$\{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$

(h) $R_3 \oplus R_6$

$\{(2,1), (3,1), (3,2)\}$

(i) $R_2 \circ R_1$

$\{(2,1), (3,1), (3,2)\}$

(j) $R_6 \circ R_6$

$\{(1,1), (1,2), (2,2), (2,3), (3,1), (3,3)\}$

Q10) (i)

$$M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(ii)

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(iii)

$$M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(iv)

$$M_R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q11) (i)

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

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(ii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $R = \{(1,2), (2,2), (3,2)\}$

(iii) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $R = \{(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)\}$

Q12)

$l(a) = l(a)$ so aRa hence R is reflexive.

Lets say aRb then $l(a) = l(b)$ so $l(b) = l(a)$

so bRa, hence R is symmetric.

Supposing aRb and bRC; $l(a) = l(b)$, $l(b) = l(c)$

so $l(a) = l(c)$.

Hence aRc so R is transitive.

R is an equivalence

Q13) (1)

$R = \{(Ali, DB), (Ali, AI), (Bilal, AI), (Bilal, OS), (Cathy, DB), (Cathy, DBS), (David, DS), (David, OS)\}$

(2)

Domain(R) = {Ali, Bilal, Cathy, David}

Range(R) = {DB, AI, DS, OS}

(3)

No because each student has 2 outputs.

(4)

reversing every order pair in R.

$R^{-1} = \{(DB, Ali), (AI, Ali), (AI, Bilal), (OS, Bilal), (DB, Cathy), (DS, Cathy), (DS, David), (OS, David)\}$

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Q14)

$F(0) = \frac{9}{5}(0) + 32 = \boxed{32°F}$ $F(25) = \frac{9}{5}(25) + 32 = \boxed{77°F}$

$F(100) = \frac{9}{5}(100) + 32 = \boxed{212°F}$

Let $F(C_1) = F(C_2)$

$\frac{9}{5}(C_1) + 32 = \frac{9}{5}(C_2) + 32$

$\boxed{C_1 = C_2 \quad \text{so } F \text{ is one-to-one.}}$

$F^{-1} = ?$

Let $y = \frac{9}{5}C + 32$ $\quad C = \frac{9}{5}y + 32$

$\frac{5(C-32)}{9} = y$

$\boxed{F^{-1}(x) = \frac{5(x-32)}{9}}$

95°F to C = ?

$F^{-1}(95) = \frac{5(95-32)}{9} = \boxed{35°C}$

Q15)

• Reflexive

$a \in Z$

$a - a = 0$

as $m \mid 0$, $a \equiv a \pmod m$

so R is reflexive.

• Symmetric

let $a \equiv b \pmod m$ then

$m \mid (a-b)$

so $a - b = km$ for some $k \in Z$

$\times -1 \Rightarrow b - a = -km$

As $-k \in Z$, $m \mid (b-a)$ so

$b \equiv a \pmod m$

so R symmetric

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• Transitive

let $a \equiv b \pmod m$, $b \equiv c \pmod m$ then

$a - b = km$ $\quad b - c = lm$

for some $k, l \in Z$

$$(a-b) + (b-c) = km + lm$$

$$a - c = (k+l)m$$

As $k+l \in Z$, $m \mid (a-c)$ so; $a \equiv c \pmod m$

hence R is transitive.

Ans: R is an equivalence relation on Z.

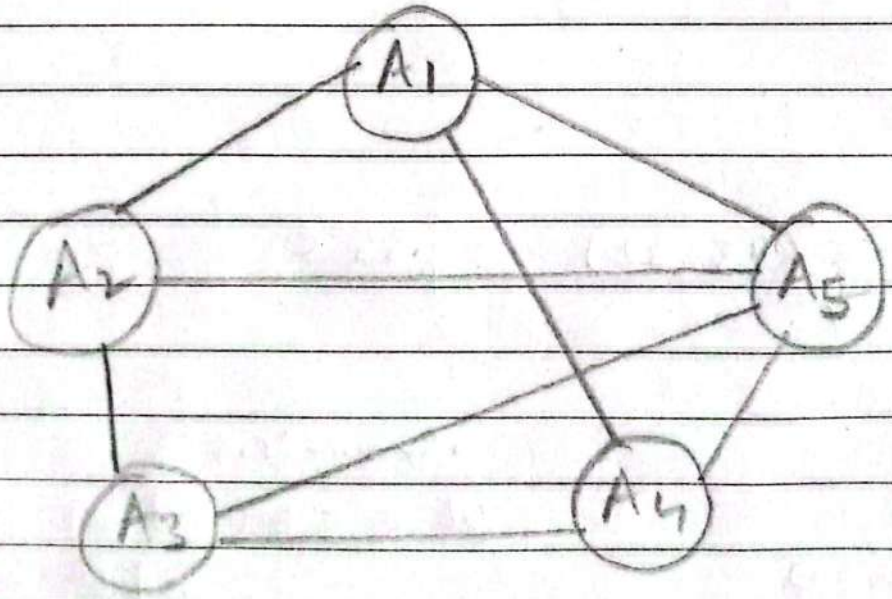
Q16)(a)

$A_1 \cap A_2 = \{0, 2, 4\}$ $\quad A_1 \cap A_5 = \{0, 8\}$

$A_1 \cap A_4 = \{6, 8\}$ $\quad A_2 \cap A_3 = \{1, 3\}$

$A_2 \cap A_5 = \{0, 1\}$ $\quad A_3 \cap A_4 = \{5, 7, 9\}$ $\quad A_3 \cap A_5 = \{1, 9\}$

$A_4 \cap A_5 = \{8, 9\}$



(b)

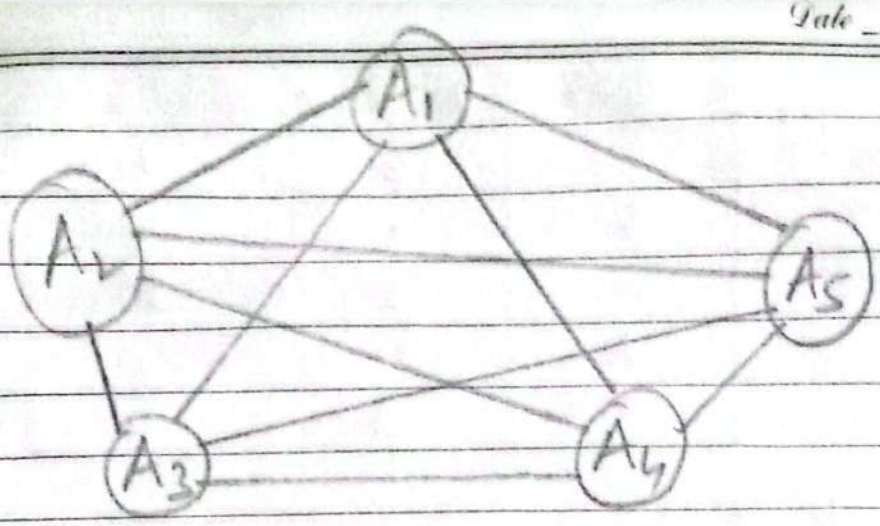
$E = \{(A_1, A_2), (A_1, A_3), (A_1, A_4), (A_1, A_5),$

$(A_2, A_3), (A_2, A_4), (A_2, A_5), (A_3, A_5),$

$(A_4, A_5)\}$

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Q17)(i)

Vertices: 5

loops (a & c): 2 edges

a and b: 3 edges d and c: 3 edges

a-e, e-d, e-b, b-d, b-c: 5 edges

2 + 3 + 3 + 5 = 13 edges

Degrees:

a: 2+3+1=6 c: 2+3+1=6 e: 1+1+1=3

b: 3+1+1+1=6 d: 3+1+1=5

Neighbour Vertices: $N(a) = \{a, b, e\}$ $N(b) = \{a, c, d, e\}$

$N(c) = \{b, c, d\}$ $N(d) = \{b, c, e\}$

$N(e) = \{a, b, d\}$

(ii)

Vertices: 9 Edges: 12

Degrees:

a: 3 b: 2 c: 4 d: 0 f: 0 e: 6

g: 4 h: 2 i: 3

Neighbour Vertices: $N(a) = \{c, e, i\}$ $N(b) = \{h, e\}$

$N(c) = \{a, e, g, i\}$ $N(d) = \emptyset$ $N(f) = \emptyset$

$N(e) = \{a, b, c, g\}$ $N(g) = \{c, e\}$

$N(h) = \{b, i\}$ $N(i) = \{a, c, h\}$

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Q18) (i)

Vertices: 5

Edges: 13

Vertex	In-degree	Out-degree
a	6	1
b	1	5
c	2	5
d	4	2
e	0	0

(ii)

Vertices: 4

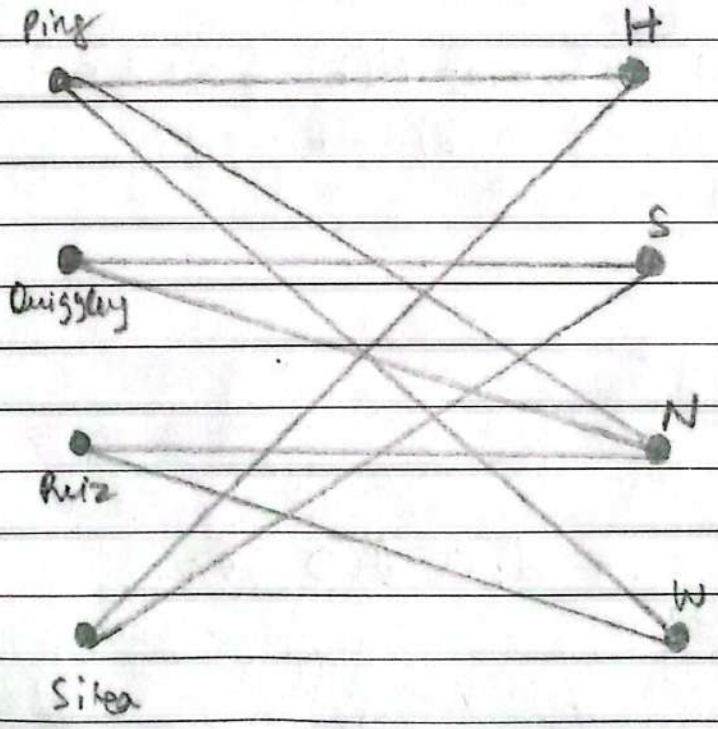
Edges: 8

Vertex	In-degree	Out-degree
a	2	2
b	3	4
c	2	1
d	1	1

Q19)

~~Employees~~: Ping → H, N, W Quigley → S, N

Ruiz → N, W Sitea → H, S



Q20) (a)

Not bipartite as b, c, f, b form cycle of 3.

(b)

Bipartite

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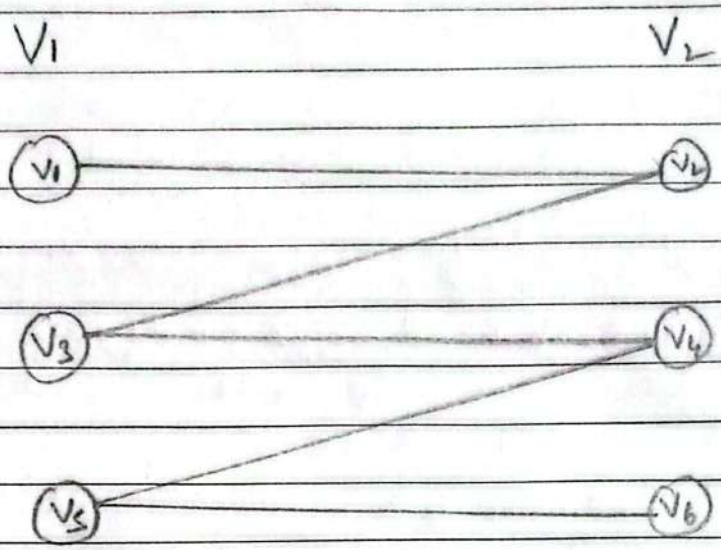
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Its a path from $v_1 - v_2 - v_3 - v_4 - v_5 - v_6$

Disjoint sets: $V_1 = \{v_1, v_3, v_5\}$

$V_2 = \{v_2, v_4, v_6\}$



(c)

Not bipartite as vertices v_2, v_4, v_5 form cycle of 3.

$v_2 \to v_4 \to v_5 \to v_2$. Odd cycle.

(d)

Not bipartite as vertices b, d, e form cycle of 3. Odd cycle.

$b \to d \to e \to b$

Q21) (G) Bipartite because vertices can be divided in 2 disjoint sets. $V_1 = \{u, w\}$ $V_2 = \{v, t, x\}$

No odd cycle.

(H) Not bipartite as contains triangle formed by b, c, e.

$b \to c \to e \to b$. Odd cycle.

(A) Not bipartite as vertices can be divided into $V_1 = \{d\}$, $V_2 = \{a, g, h, i, j, k, f\}$. No. Has odd cycle of 7.

$u \to v \to w \to x \to y \to z \to t \to u$.

(B) Bipartite as it can be divided in 2 disjoint sets.

$V_1 = \{d\}$, $V_2 = \{f, g, h, i, j, k, l\}$.

No odd cycles.

Q22) (i)

1st graph: Hexagon

$E_1 = \{af, ab, bc, cd, de, ef\}$

2nd graph: Triangle. $\rightarrow E_2 = \{fb, bd, df\}$

Union: $V = \{a, b, c, d, e, f\}$

$E = \{af, ab, bc, cd, de, ef, fb, dd, df\}$ Ans

(ii)

1st graph: $\rightarrow E_1 = \{ae, be, ce, de\}$

2nd graph: $\rightarrow E_2 = \{af, fb, fe, eg, ec, gd\}$

Union: $V = \{a, b, c, d, e, f, g\}$

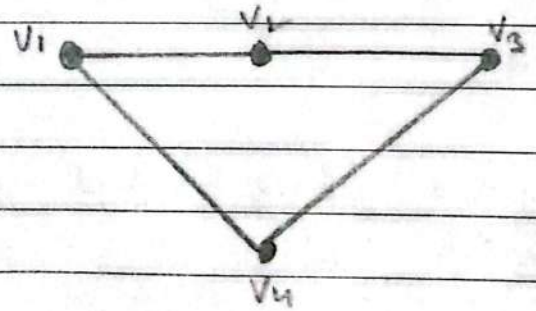
$E = \{ae, be, ce, de, af, fb, fe, eg, ec, gd\}$ Ans.

Q23) (a)

Sum of degrees $= 1+1+2+3 = 7$, ~~even~~ odd so no graph exists.

(b)

Sum of degrees $= 1+1+3+3 = 8$, even so graph exists.



(c)

In simple graph with 4 vertices, vertex of degree 3 must be connected to all other 3 vertices. So if 2 vertices have degree 3, remaining 2 vertices must be connected to both of them, giving each of the remaining vertices a degree of atleast 2 not 1. So no such simple graph exists.

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(Q24)(a)

$\sum deg(v) = 2|E|$

If each of 15 have 3 friends,

$15 \times 3 = 45$

Sum of degrees must be even because it is equal to twice the no. of edges. As 45 is odd, NO its not possible.

(b)

$deg(v) = 3$ for every vertex.

total edges $= |E| = \frac{4 \times 3}{2} = 6$

Graph is complete K_4.

Yes it is possible.

(c)

Each handshake occurs between unique pair.

$|E| = \binom{10}{2} = \frac{10(10-1)}{2} =$ 45 handshakes.

In graph theory, each person is represented by a vertex and each handshake by an edge.

(d)

Case 1: Exactly 2 friends each.

Sum of degrees $= 5 \times 2 = 10$. Even so graph is possible.

Case 2: Exactly 3 friends each.

Sum of degrees $= 5 \times 3 = 15$. Odd so graph can't exist.

(e)

$\sum deg(v) = 2|E|$

let n: no. of vertices. every vertex's deg = 4

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$4n = 2(10)$

~~$n = 5$ vertices~~

$4n = 20$

$n = 5$ vertices

Q25) (a)

All 4 graphs are connected.

(b)

W is cycle tree.

(c)

W, Y, Z are loop free.

(d)

Graphs W and Y are simple as they have no loops/many edges.

Q26)

(a)

The graphs are isomorphic.

$g : V(G) \to V(G')$

$g(v_1) = w_2 \quad g(v_2) = w_3 \quad g(v_3) = w_1$

$g(v_4) = w_5 \quad g(v_5) = w_4$

(b)

Isomorphism exists.

$g(v_1) = u_5 \quad g(v_2) = u_2 \quad g(v_3) = u_4 \quad g(v_4) = u_3$

$g(v_5) = u_1 \quad g(v_6) = u_6$

(c)

Graphs are isomorphic.

$g(v_1) = u_2 \quad g(v_2) = u_3 \quad g(v_3) = u_4$

$g(v_4) = u_5 \quad g(v_5) = u_6 \quad g(v_6) = u_1$

$g(v_7) = u_7$

Q27)

(a)

$a \to c \to b \to i \to m \to p \to s \to z$

$4 + 2 + 2 + 2 + 2 + 2 + 2$

$= 16$

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(b)

a → c → d → e → g → z ;

d(a) = 0 d(c) = 3 d(d) = 3+3 = 6 d(b) = 4

d(e) = 6+1 = 7 d(g) = 7+5 = 12

d(z) = 12+4 = **16**

(c)

a → c → e → z

d(a) = 0 d(c) = 10 d(e) = 10+3 = 13 d(z) = 13+6 = **19**

Q28) (a)

AB = 30 BC = 30 CD = 25 DA = 40 AC = 50 BD = 35

Ans **A → B → C → D → A : 30 + 30 + 25 + 40 = 125 km** Ans

A → B → D → C → A : 30 + 35 + 25 + 50 = 140

A → C → B → D → A : 50 + 30 + 35 + 40 = 155

(b)

A → B → C → D → A : 20 + 30 + 12 + 35 = **97 km**

A → B → D → C → A : 20 + 34 + 12 + 42 = 108

A → C → B → D → A : 42 + 30 + 34 + 35 = 141

Ans : A → B → C → D → A

(c)

A → B → C → D → A : 10 + 35 + 30 + 20 = 95

A → B → D → C → A : 10 + 25 + 30 + 15 = **80 km**

A → C → B → D → A : 15 + 35 + 25 + 20 = 95

Ans : A → B → D → C → A

Q29)

There are 6 rooms with odd interior degrees (A, B, C, D, E, G)

So its not mathematically possible to find a trail in room A that ends in room B.

Q30)

(1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5)

5 vertices, edges $= \frac{5(5-1)}{2} = 10$

$\deg(v) = 4$

As every vertex has an even degree and the graph is connected, it has an Euler circuit.

1 → 2 → 3 → 1 → 4 → 2 → 5 → 3 → 4 → 5 → 1

All 10 dominoes can be arranged in closed seior.

Q31) (i)

Circuit: $v_0 \to v_1 \to v_5 \to v_4 \to v_7 \to v_6 \to v_2 \to v_3 \to v_0$

Path: $v_0 \to v_1 \to v_5 \to v_4 \to v_7 \to v_6 \to v_2 \to v_3$

(ii)

Circuit doesn't exist as vertices b and d have degree of 2. Any circuit must enter and exit through these vertices using edges: (a,b), (b,c), (a,d), (d,c).

Path: b → a → e → h → g → f → c → d

(iii)

- Circuit and path both don't exist.
- Vertices b and f have a degree of 2 and are only
- connected to their outer ring neighbours.
- b and f are isolated on opposite ends of the outer ring.

Q32) (i)

Circuit: a → b → c → d → e → f → g → h → a

Path: a → b → c → d → e → f → g → h

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(ii)

Circuit doesn't exist as degree of 2 forces conflicting edge requirements at central vertex j.

Path: ~~a → b → c → j → e → f~~

h → a → b → c → j → g → f → e → d

(iii)

Circuit : x → a → y → b → z → c → x

Path : x → a → y → b → z → c

Q33) (i)

Exists. $v_1 \to v_2 \to v_5 \to v_2 \to v_3 \to v_4 \to v_5 \to v_4 \to v_1$

(ii)

Doesn't exist. because graph has odd degree vertices v_3, v_7

(iii)

Graph contains vertices of odd degree so it doesn't have an euler circuit.

Q34) (i)

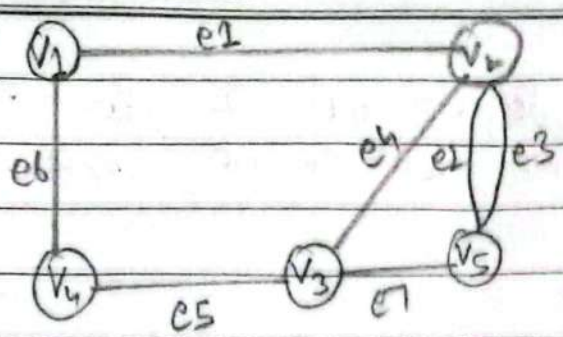
No doesn't have an euler path. Graph contains 4 vertices with odd degrees (u, e, h, w).

(ii)

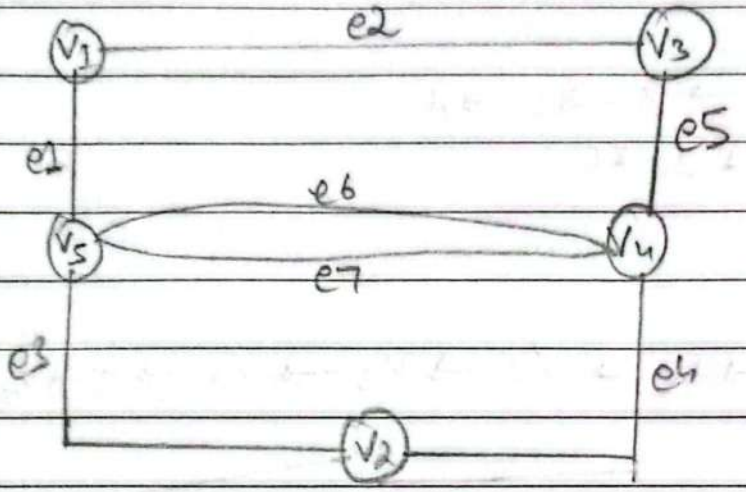
Yes euler path exists. Path : ~~u → v₀~~ u → $v_0 \to v_1 \to$ u → $v_7 \to v_0$

Q35) (i)

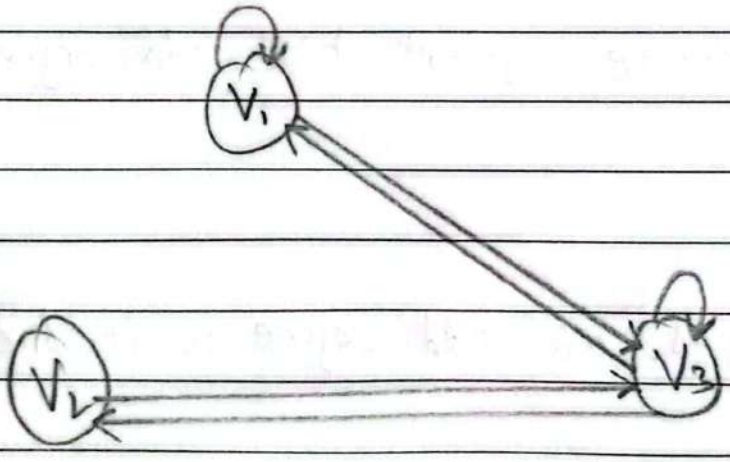
Date _____



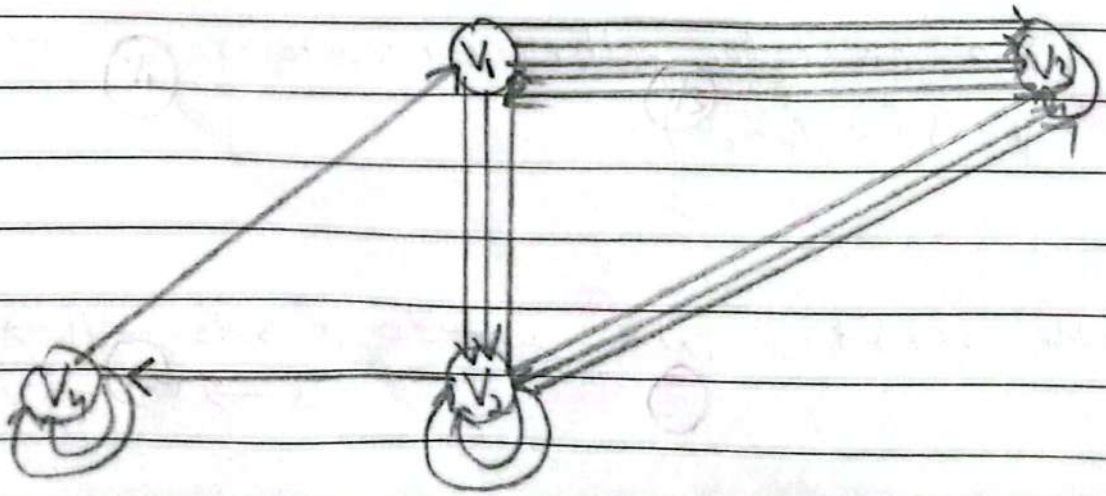
(ii)



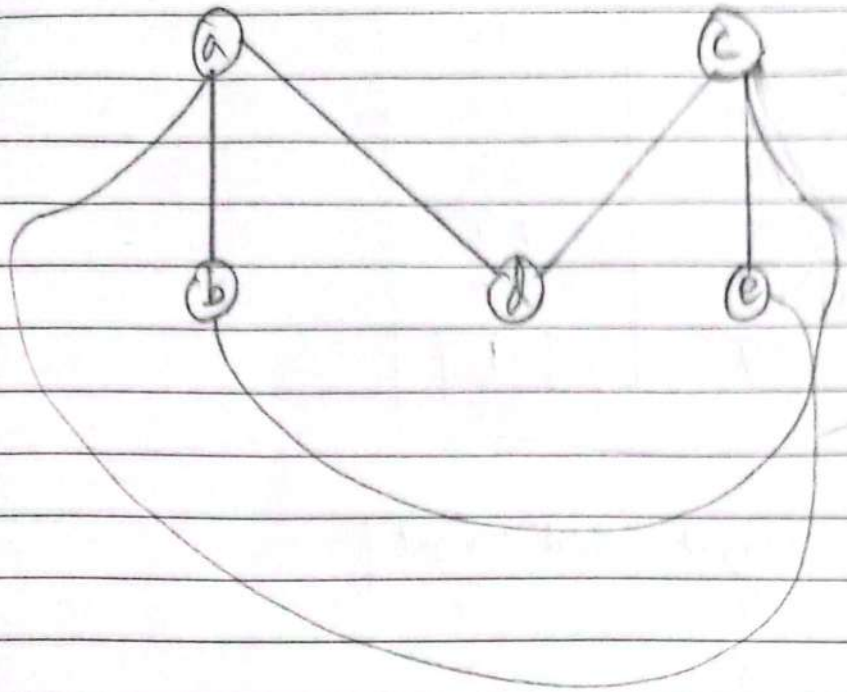
(26)(i)



(ii)



Q37) (i)
Yes its planar. Complete bipartite graph.



(ii)
Not planar.

Q38) Euler Formula : $V - E + R = 2$

(G)

Vertices : 4 (V) Edges : 6 (E) Regions : 4 (R)

$V - E + R = 4 - 6 + 4 = 2$

Verified

(H)

V : 6 E : 9 R : 5

$V - E + R = 6 - 9 + 5 = 2$ Verified

(I)

V : 5 E : 10 R : 7

$V - E + R = 5 - 10 + 7 = 2$ verified

Q39) (a) level 3
(b) 0
(c) 5
(d) u, v
(e) d
(f) K and L
(g) m, s, t, x, y
(h) a, b, c, d, e, f, h, i, k, m, n, o, t, v
(i) v, n, h, d, a
(j) g, j, l, p, q, r, s, u, w, x, y, z

Date

Q40)(i)

Edge added	Weight	Total weight
$V_0 - V_5$	4	4
$V_5 - V_6$	8	12
$V_5 - V_4$	10	22
$V_4 - V_3$	2	24
$V_3 - V_1$	5	29
$V_6 - V_7$	13	42
$V_3 - V_2$	19	61

(ii)

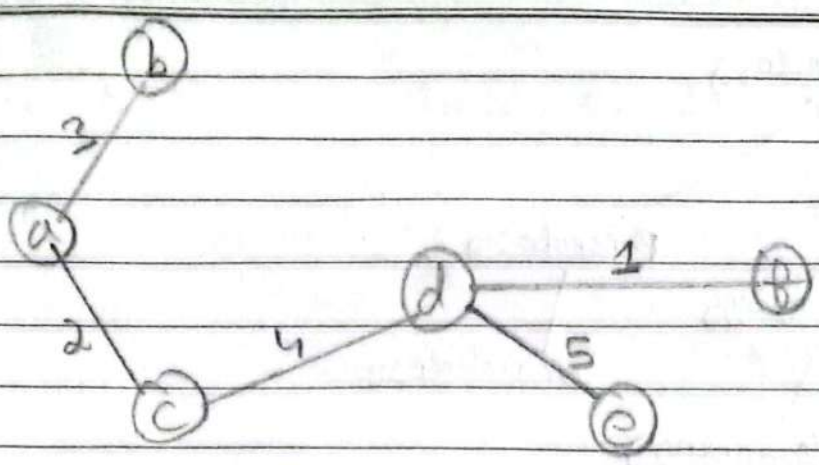
Edge Added	Weight	Total Weight
$V_0 - V_1$	4	4
$V_0 - V_8$	8	12
$V_8 - V_7$	1	13
$V_7 - V_6$	2	15
$V_6 - V_3$	4	19
$V_3 - V_2$	2	21
$V_3 - V_4$	7	28
$V_4 - V_5$	9	37

Q41)(i)

Edge ~~Added~~ selected	Weight	
df	1	
ac	2	
ab	3	
bc	~~7~~ -	→ Rejected
cd	~~8~~ 4	
de	5	

$w = 1+2+3+4$

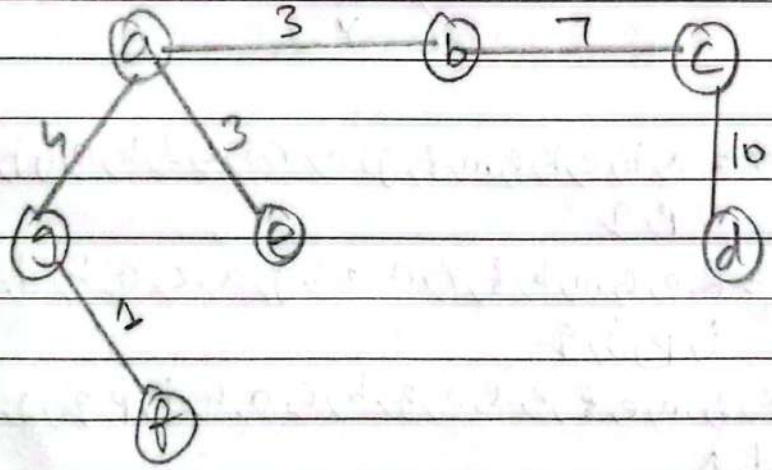
$w = 15$



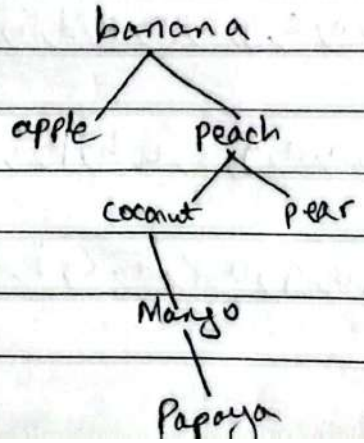
(ii)

Edges selected	Weight	
gf	1	
ab	3	
ae	3	
ag	4	
be	–	→ Rejected
ec	7	
cd	10	

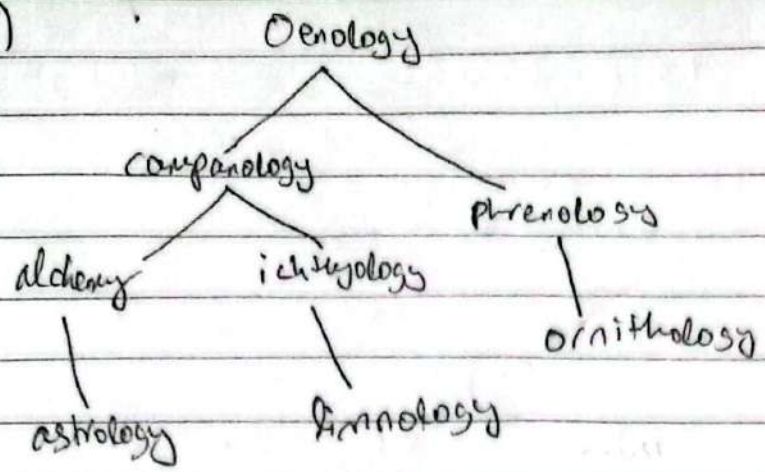
$W = 1+3+3+4+7+10 = \boxed{28}$



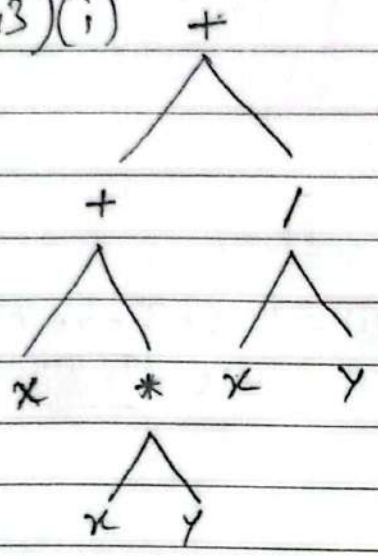
Q42) (i)



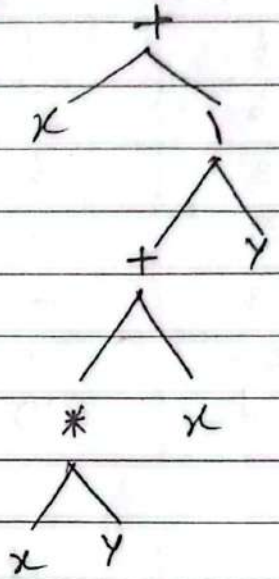
(ii)



Q43) (i)



(ii)



Q44) (i)

~~Post~~ • PreOrder Traversal: Order = a, b, e, k, l, m, f, g, n, r, s, c, d, h, o, i, j, p, q

• InOrder Traversal: Order = k, e, l, m, b, f, r, n, s, g, a, c, o, h, d, i, p, j, q

• PostOrder Traversal: Order = k, l, m, e, f, r, s, n, g, b, c, o, h, i, p, q, j, d, a

(ii)

• PreOrder Traversal: Order = a, b, e, f, l, m, n, c, g, h, o, p, d, i, j, q, s, t, u, r, k

• InOrder Traversal: Order = e, b, l, f, m, n, a, g, c, o, h, p, i, d, s, q, t, u, j, r, k

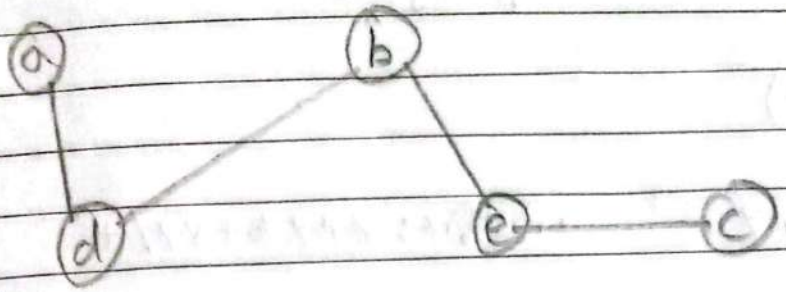
PostOrder Traversal: Order = e, l, m, n, f, b, g, o, p, h, c, i, s, u, t, q, r, j, k, d, a

CamScanner

Q45)(i)

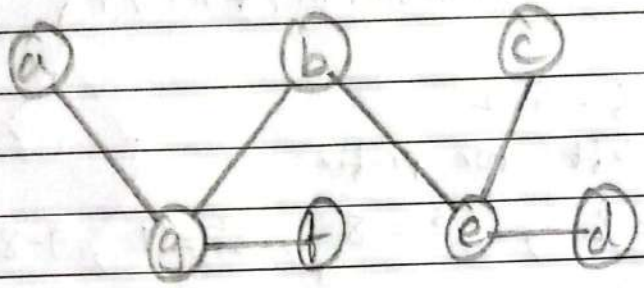
Removed edges: ab which removes cycle a–b–d–a

: bc which removes cycle b–c–e–b



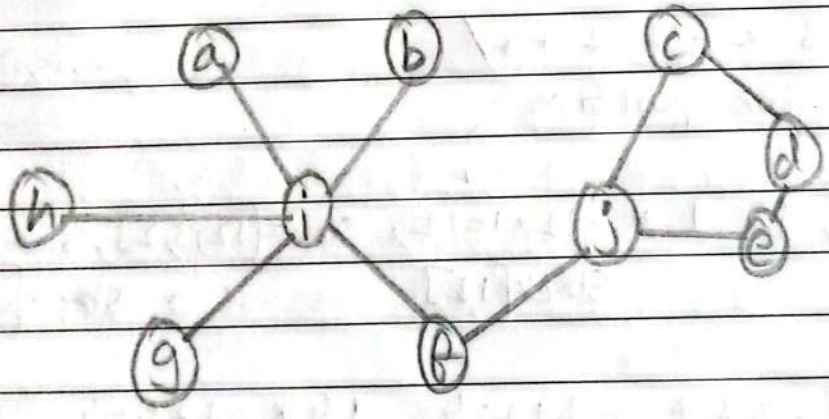
(ii)

Removed edges: ad, ge, ab, bc, cd, bf, fe, de



(iii)

Removed edges: ha, ab, bc, cd, de, ef, fg, gh, bi, fj, ij, jd



Q46)

(a) Edges = n−1

= 10000 − 1 = 9999

Ans

(b) V = ni + 1

= 2(1000) + 1 = 2001

Ans

Edges = 2001 − 1

= 2000

Ans

(c) V = ni + 1

= 5(100) + 1

= 501 Ans

(d) $V = mi + 1$, $L = V - i$ → $1000 = 3i + 1$, $i = 33$ internal vertices

$L = 100 - 33 = \boxed{67}$ Ans

Q47)

(i) $(x + xy) + (x/y)$

• Prefix: $++x*xy/xy$ • Postfix: $xxy*+xy/+$

(ii) $x + ((xy + x)/y)$

• Prefix: $+x/+*xyxy$ • Postfix: $xxy*x+y/+$

Q48) (i) $+-\uparrow 3\ 2\uparrow 2\ 3/6-4\ 2$

right to left for prefix

$4-2=2$, $6/2=3$, $2^3=8$, $3^2=9$, $9-8=1$, $1+3=\boxed{4}$

$+-\uparrow 3\ 2\uparrow 2\ 3/6\ 2$ ⟵ $+-\uparrow 3\ 2\uparrow 2\ 3\ 3$ ⟵ $+-\uparrow 3\ 2\ 8\ 3$ ⟵ $+1\ 3$ Ans

(ii) $4\ 8 + 6\ 5 - * 3\ 2 - 2\ 2 + */$

left to right for postfix

$4+8=12$, $6-5=1$, $12\ 1\ *=12\times1=12$, Stack[12,3,2], $3-2=1$,

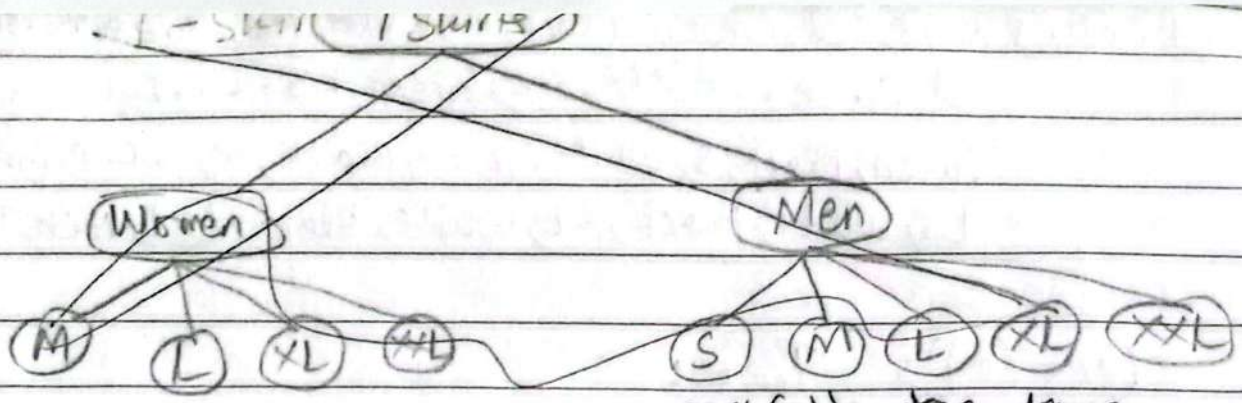
Stack[12,1] Stack[12] Stack[12,1]

Stack[12,1,2,2], $2\ 2+ = 2+2=4$, $1\ 4\ * = 1*4=4$,

Stack[12,1,4] Stack[12,4]

$12\ 4\ / = 12/4 = \boxed{3}$ Ans

Q49) (a)



part (a)'s tree drawn after part d.

Women's total = 4+4+3+2 = 13

Men's total = 3+3+3+2+2 = 13

Total t shirts = 13+13 = **26** Ans

(b)

No. of internal vertices:

1 root

2 gender branches

Size vertice = 5+4 = 9

Internal vertices = 1+2+9 = 12

No. of leaves = 26

Height of tree:

Root → Gender → Size → Color

0 1 2 3

Height = 3

(c)(i) No its not a full m-ary tree as the internal vertices do not all have the same constant number of children (m).

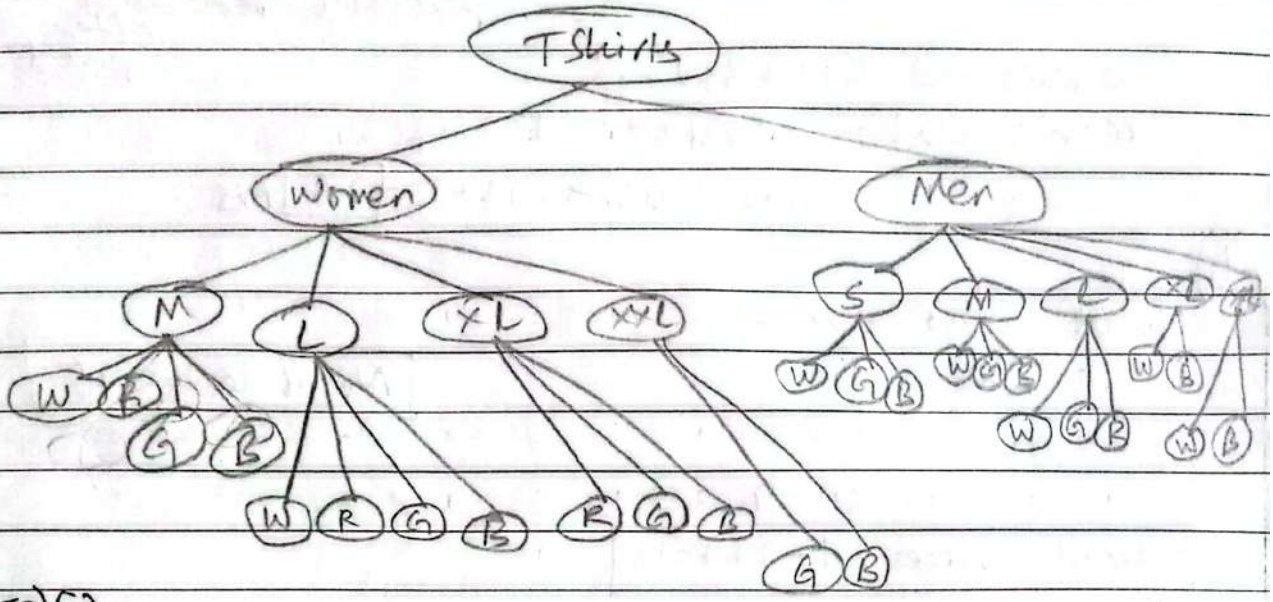
(ii) Yes because all leaves reside at the exact same depth (level = 3).

(d) • PreOrder: TShirts, Women, M, White, Red, Green, Black, L, White, Red, Green, Black, XL, Red, Green, Black, XXL, Green, Black, Men, S, White, Green, Black, M, White, Green, Black, L, White, Green, Black, XL, White, Black, XXL, White, Black

Date _____

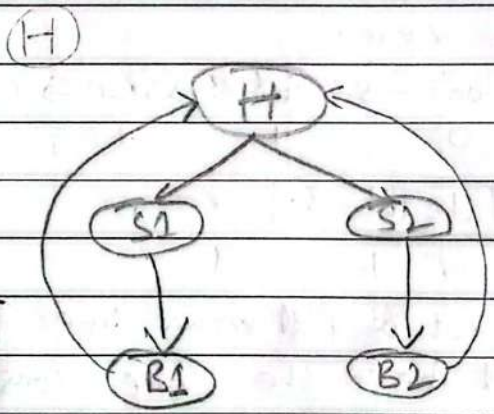
- Post Order: White, Red, Green, Black, M, White, Red, Green, Black, L, Red, Green, Black, XL, Green, Black, XXL, Women, White, Green, Black, S, White, Green, Black, M, White, Green, Black, L, White, Black, XL, White, Black, XXL, Men, TShirts

Part (a)'s Tree:



Q50)(a)

H: head of platoon
S1 & S2: soldier 1 & 2
B1 & B2: Subordinate 1 & 2



(a)

Vertex	Indegree	Outdegree
H	2	2
S_1	1	1
S_2	1	1
B_1	1	1
B_2	1	1

(b)

- Equivalence Relation:

-Not reflexive: no self loops
- Not symmetric: $H \rightarrow S_1$ but $S_1 \not\rightarrow H$
-Not transitive: $H \rightarrow S_1$ & $S_1 \rightarrow B_1$ but $H \not\rightarrow B_1$

- Partial Order Relation

-Not reflexive: no self loops
-Not antisymmetric: $H \rightarrow S_1 \rightarrow B_1 \rightarrow H$.
-Not transitive: $H \rightarrow S_1$, ~~$B_1 \rightarrow S_1 \rightarrow$~~ B_1 but $H \not\rightarrow B_1$

So there's no equivalence relation nor a partial order.

Q5 1) (a)

Iteration	Visited vertex	d(A)	d(B)	d(C)	d(D)	d(E)	d(F)	d(School)
Init	-	∞	∞	∞	∞	∞	∞	∞
1	Home	3	2	5	3	∞	∞	∞
2	B	3	2	5	3	∞	∞	∞
3	A	3	2	5	3	∞	∞	∞
4	D	3	2	5	3	7	7	∞
5	C	3	2	5	3	7	7	∞
6	E	3	2	5	3	7	7	∞
7	F	3	2	5	3	7	7	9
8	School	3	2	5	3	7	7	9

Shortest Path: Home → B → D → F → School

Edge Weights: 2 + 1 + 4 + 2 = 9

Ans

(b)

- Degree of each vertex

deg(Home) = 3 deg(C) = 2

deg(A) = 2 deg(D) = 3 deg(F) = 3

deg(B) = 3 deg(E) = 4 deg(School) = 2

Date

- Odd degree vertices: 4 • Even degree vertices: 4
- No euler circuit exists because this graph has 4 odd vertices.
- No euler path exists because 4 odd vertices exist and for euler path there must be 0 or exactly 2 odd vertices.

(c)

H. Path: Every vertex visited once, H. Circuit: Every vertex visited once and returns to starting vertex

Let total vertices (V) = 8

To form a H. Circuit we must enter and leave every vertex using 2 edges or ~~1~~ and return to start.

To visit A → (Home, A), To visit C → (Home, C) and (C, E)

No capacity ~~for~~ for Home to connect to B.

So no Hamilton circuit exists.

For Hamilton Path:

~~Home → C → E → B → A → D → F → School~~

C → Home → A → D → B → E → F → School

All 8 vertices visited

so h. path exists.